

Module 6 | Simultaneous Nonlinear Equations and Basins of Attraction



In the last chapter, we studied stationary iterations for linear systems, and we used the fixed-point viewpoint together with norm estimates to decide when an iteration converges. In the final two lectures, we maintain the same theoretical framework, but the target problem is now nonlinear:

$$f(\vec{x}) = \vec{0}, \quad f : \mathbb{R}^n \rightarrow \mathbb{R}^n.$$

Even when $f(\vec{x}) = \vec{0}$ has multiple solutions, an iterative method can converge to different solutions depending on the initial guess. The resulting global picture is described using *basins of attraction*.

§6.1 Simultaneous Iteration

Similar to the last chapter, we rewrite the root-finding problem as a fixed-point equation,

$$f(\vec{x}) = \vec{0} \iff g(\vec{x}) = \vec{x},$$

and apply the iteration sequence

$$\vec{x}^{(k+1)} = g(\vec{x}^{(k)}).$$

Because there is no canonical choice of g , different formulations will exhibit varied convergence behavior. We require a convergence criterion that is (i) local, (ii) computationally checkable, and (iii) aligns with our established linear theory.

6.1.1 Local Convergence and the Jacobian Test

Definition 6.1.38: Jacobian matrix

Let $g = (g_1, \dots, g_n)^T$ and assume all first partial derivatives exist at $\vec{x} \in \mathbb{R}^n$. The *Jacobian matrix* of g at $\vec{x}$ is

$$J_g(\vec{x}) = \left[\frac{\partial g_i}{\partial x_j}(\vec{x}) \right]_{i,j=1}^n.$$

The following theorem is the nonlinear analogue of the contraction criterion for linear iterative methods.

Theorem 6.1.39: Jacobian test for local convergence

Let $D \subset \mathbb{R}^n$ be closed and let $g : D \rightarrow D$ be continuous. Assume $\vec{\xi} \in D$ is a fixed point: $g(\vec{\xi}) = \vec{\xi}$. Assume the entries of J_g exist and are continuous in a neighborhood of $\vec{\xi}$ contained in D . If

$$\|J_g(\vec{\xi})\|_\infty < 1,$$

then there exists $\varepsilon > 0$ such that $g(\overline{B_\varepsilon(\vec{\xi})}) \subset \overline{B_\varepsilon(\vec{\xi})}$ and, for every $\vec{x}^{(0)} \in \overline{B_\varepsilon(\vec{\xi})}$, the iteration $\vec{x}^{(k+1)} = g(\vec{x}^{(k)})$ converges to $\vec{\xi}$.

Remark. If $g(\vec{x}) = T\vec{x} + \vec{c}$ is an affine map (the exact linear case analyzed in last chapter), then $J_g(\vec{x}) = T$ for all $\vec{x}$. Consequently, [theorem 39](#) directly recovers the linear convergence criterion: local linearization effectively replaces the constant iteration matrix.

Proof of [theorem 39](#).

Continuity of J_g gives a small closed ball around ξ where $\|J_g(\vec{x})\|_\infty$ is uniformly < 1 . For $\vec{x}$ and $\vec{y}$ in such a ball, use the mean value theorem along the line segment from $\vec{x}$ to $\vec{y}$ to bound $\|g(\vec{x}) - g(\vec{y})\|_\infty$ by

$$\|g(\vec{x}) - g(\vec{y})\|_\infty \leq \max_{t \in [0,1]} \|J_g(t\vec{x} + (1-t)\vec{y})\|_\infty \|\vec{x} - \vec{y}\|_\infty.$$

So g is a contraction on that ball. The result follows the contraction mapping theorem in $\mathbb{R}^n$. ■

■ Question 73.

Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$g(x_1, x_2) = \left(\frac{1}{2} \cos x_2, \frac{1}{2} \sin x_1 \right).$$

- (a) Compute $J_g(\vec{0})$ and $\|J_g(\vec{0})\|_\infty$. What does [theorem 39](#) guarantee for the iteration $\vec{x}^{(k+1)} = g(\vec{x}^{(k)})$ near $\vec{0}$?
- (b) Run a few numerical steps starting from $\vec{x}^{(0)} = (1, 1)^T$. What behavior do you observe?

6.1.2 Multidimensional Relaxation

A simple, initial choice for an iterative map is

$$\vec{x}^{(k+1)} = g_\lambda(\vec{x}) = \vec{x} - \lambda f(\vec{x}), \quad \lambda > 0,$$

The scalar λ is called the *relaxation parameter*. If the sequence $(\vec{x}^{(k)})$ converges to a limit $\vec{\xi} \in \mathbb{R}^n$ and f is continuous near $\vec{\xi}$, passing to the limit shows that $f(\vec{\xi}) = \vec{0}$.

What does λ do? Rather than taking a full unscaled step, it scales the correction vector before applying it.

- **Under-Relaxation** ($0 < \lambda \ll 1$): If an iteration overshoots the root $\vec{\xi}$, resulting in oscillations or divergence, a small λ keeps the new iterate close to $\vec{x}^{(k)}$, taming the oscillations.
- **Over-relaxation** ($\lambda > 1$): If the iteration creeps toward the solution too slowly, a larger λ accelerates progress along the correction direction.

The primary difficulty lies in selecting an optimal λ .

Because $g_\lambda(\vec{x}) = \vec{x} - \lambda f(\vec{x})$, at the root $\vec{\xi}$ where $f(\vec{\xi}) = \vec{0}$, we have

$$J_{g_\lambda}(\vec{\xi}) = I - \lambda J_f(\vec{\xi}).$$

Writing the error at the k th step as $\vec{e}^{(k)} = \vec{x}^{(k)} - \vec{\xi}$, a Taylor expansion of f around $\vec{\xi}$ gives

$$f(\vec{\xi} + \vec{e}) = J_f(\vec{\xi})\vec{e} + O(\|\vec{e}\|^2),$$

so the error recursion becomes

$$\vec{e}^{(k+1)} = \underbrace{(I - \lambda J_f(\vec{\xi}))}_{=J_{g\lambda}(\vec{\xi})} \vec{e}^{(k)} + O(\|\vec{e}^{(k)}\|^2).$$

Thus the iteration converges locally when $I - \lambda J_f(\vec{\xi})$ is a contraction, i.e. when $\rho(I - \lambda J_f(\vec{\xi})) < 1$. The following theorem gives a sufficient hypothesis under which such a λ can be found.

Theorem 6.1.40: Local convergence under diagonal dominance

Assume $f(\vec{\xi}) = \vec{0}$. Assume the first partial derivatives of f exist and are continuous near $\vec{\xi}$. If $J_f(\vec{\xi})$ is diagonally dominant, then there exist $\varepsilon > 0$ and $\lambda > 0$ such that the iteration $\vec{x}^{(k+1)} = \vec{x}^{(k)} - \lambda f(\vec{x}^{(k)})$ converges to $\vec{\xi}$ for all $\vec{x}^{(0)} \in \overline{B_\varepsilon(\vec{\xi})}$.

Proof of theorem 40.

[Proof sketch] Let $A = J_f(\vec{\xi})$. Strict diagonal dominance implies that A is nonsingular (Why?).

We want all eigenvalues μ_j of $I - \lambda A$ satisfy $|\mu_j| < 1$. Note that $\mu_j = 1 - \lambda \sigma_j$ where σ are eigenvalues of A . Choosing

$$\lambda = \frac{1}{\max_i a_{ii}}$$

forces $|1 - \lambda \sigma_j| < 1$ for each j . ■

The condition of strict diagonal dominance will only be satisfied in a small class of problems (although this class does contain some examples of practical importance). More generally, if we want to minimize the norm of the Jacobian, we want to replace λ by a matrix Λ .

6.1.3 Newton's Method for Systems

If we replace the scalar λ by a constant matrix Λ , a natural generalised iteration is given by

$$\vec{x}^{(k+1)} = \vec{x}^{(k)} - \Lambda f(\vec{x}^{(k)}).$$

The optimal local choice would naturally be $\Lambda = [J_f(\vec{\xi})]^{-1}$, which forces $J_g(\vec{\xi}) = 0$ and hence guarantees superlinear convergence near the root. But a priori, $\vec{\xi}$ is unknown. So, Newton's method uses the Jacobian of the current iterate as an approximation.

Definition 6.1.41: Newton's method for $f(\vec{x}) = \vec{0}$

Assume $J_f(\vec{x}^{(k)})$ exists and is nonsingular for each iterate. The Newton's method recursion is

defined by

$$\vec{x}^{(k+1)} = \vec{x}^{(k)} - [J_f(\vec{x}^{(k)})]^{-1} f(\vec{x}^{(k)}), \quad k = 0, 1, 2, \dots$$

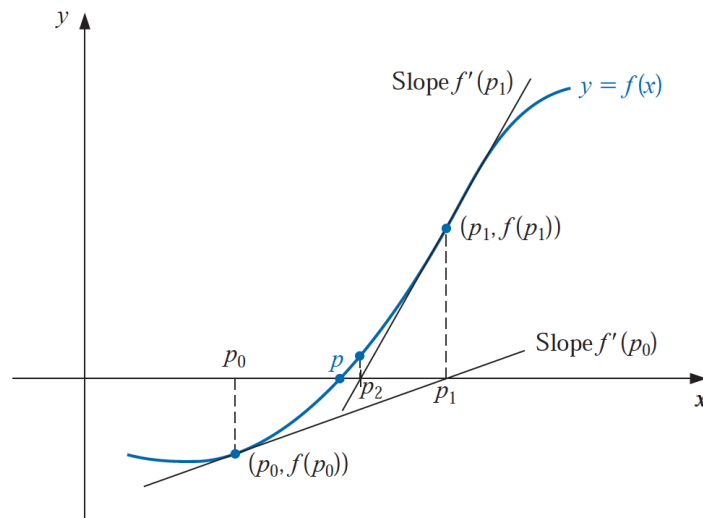


Figure 6.1: Newton's Method in One Dimension

In practice, we never explicitly form the inverse. Each Newton step requires solving one linear system: find $\vec{\Delta x}^{(k)}$ satisfying

$$J_f(\vec{x}^{(k)}) \vec{\Delta x}^{(k)} = -f(\vec{x}^{(k)}),$$

then set $\vec{x}^{(k+1)} = \vec{x}^{(k)} + \vec{\Delta x}^{(k)}$. This inner step may be performed using Gaussian elimination or iterative methods (Jacobi or Gauss-Seidel) from the last chapter.

■ Question 74.

Group

Given a fixed $c > 0$:

(a) For what simple function $f(x)$ will Newton's method estimate $\frac{1}{c}$ without using its value?

(Note $f(x) = \frac{1}{c} - x$ doesn't work because it requires us to know the value $\frac{1}{c}$ beforehand.)

(b) For what simple function $f(x)$ will Newton's method estimate $\sqrt{c}$ without using its value?

(c) In both cases, graph f to decide which original zero-estimates x_0 will work (i.e., cause the method to converge to the desired zero). Not all initial estimates work.

6.1.4 Broyden's Method

Newton's method has a structural cost that grows quickly: evaluating all n^2 partial derivatives of $J_f(\vec{x}^{(k)})$ at each step costs $O(n^2)$ function evaluations, compared to $O(n)$ for a single evaluation of f . A more subtle approximation is thus to evaluate the Jacobian operator only approximately, even if the resulting approximate linear system is solved exactly.

Broyden's method is motivated by asking: *can we avoid recomputing the full Jacobian at every step, and instead update our current approximation cheaply using only information we already have?*

A Generalization of Secant Method

In one dimension, we could replace $f'(x^{(k)})$ by a finite-difference approximation

$$f'(x^{(k)}) \approx \frac{f(x^{(k)}) - f(x^{(k-1)})}{x^{(k)} - x^{(k-1)}},$$

This is called the secant method.

In n dimensions, the analogous idea is to replace $J_f(\vec{x}^{(k)})$ by a matrix B_k such that

$$B_k \cdot (\vec{x}^{(k)} - \vec{x}^{(k-1)}) = f(\vec{x}^{(k)}) - f(\vec{x}^{(k-1)}) \quad (\text{Condition 1})$$

Our goal is to find the B_k with minimal effort possible. In particular, we want to consider how to construct B_k from B_{k-1} while minimizing $\|B_k - B_{k-1}\|$.

Note that the only direction in which we *do* have fresh information about f is the step direction $\vec{\Delta x}^{(k-1)} = \vec{x}^{(k)} - \vec{x}^{(k-1)}$, where we observed that f changed by $\vec{\Delta f}^{(k-1)} = f(\vec{x}^{(k)}) - f(\vec{x}^{(k-1)})$ along this direction.

Every vector in $\mathbb{R}^n$ can be written as the sum of a multiple of $\vec{\Delta x}^{(k-1)}$ and a vector in its orthogonal complement.

We have *no* information about how f changes in directions orthogonal to $\vec{\Delta x}^{(k-1)}$, so the easiest choice is to leave B_{k-1} unchanged in that direction

$$B_k \cdot \vec{v} = B_{k-1} \cdot \vec{v} \quad \text{whenever } (\vec{\Delta x}^{(k-1)})^T \vec{v} = 0. \quad (\text{Condition 2})$$

■ Question 75.

□

Think about what [Condition 1](#) and [Condition 2](#) together force B_k to look like. Specifically:

- (a) The difference $B_k - B_{k-1}$ must vanish on the entire orthogonal complement of $\vec{\Delta x}^{(k-1)}$. What does that tell you about the rank of $B_k - B_{k-1}$?
- (b) Given that rank, what is the most general form $B_k - B_{k-1}$ can take?
- (c) Now use [equation \(Condition 1\)](#) to solve for the free parameter.

The General Iteration

Definition 6.1.42: Broyden's method

Given an initial approximation $B_0 \approx J_f(\vec{x}^{(0)})$ (often computed by finite differences), Broyden's method generates iterates by, for $k = 0, 1, 2, \dots$:

- (a) Compute the step: solve $B_k \vec{\Delta x}^{(k)} = -f(\vec{x}^{(k)})$ and set $\vec{x}^{(k+1)} = \vec{x}^{(k)} + \vec{\Delta x}^{(k)}$.
- (b) Observe the change in f : $\vec{\Delta f}^{(k)} = f(\vec{x}^{(k+1)}) - f(\vec{x}^{(k)})$.

(c) Update the Jacobian approximation:

$$B_{k+1} = B_k + \frac{(\overrightarrow{\Delta f}^{(k)} - B_k \overrightarrow{\Delta x}^{(k)})(\overrightarrow{\Delta x}^{(k)})^T}{\|\overrightarrow{\Delta x}^{(k)}\|_2^2}.$$

Cost comparison. Each Broyden step requires one evaluation of f (to compute $\overrightarrow{\Delta f}^{(k)}$) and one rank-one matrix update ($O(n^2)$ arithmetic operations), but zero derivative evaluations. Newton, by contrast, requires forming an $n \times n$ Jacobian ($O(n^2)$ partial derivative or function evaluations) and an $O(n^3)$ linear solve at every step. For large systems or problems where derivatives are expensive, Broyden's per-step savings are substantial. Note also that the linear solve in step (a) doesn't need to be done from scratch each time. There is a formula called the Sherman-Morrison formula that allows B_{k+1}^{-1} to be updated from B_k^{-1} in $O(n^2)$ operations, avoiding a full factorization.

§6.2 Domains of Convergence and Basins of Attraction

In the previous sections, we introduced a notion of convergence for zero-finding methods to indicate when the zero-estimates become arbitrarily close to a zero. In this section, we'll explore computational effort as a means of comparing different converging methods.

Definition 6.2.43: Order of convergence

Let $\vec{x}^{(k)} \rightarrow \vec{\xi}$ in $\mathbb{R}^n$ and set $e_k = \|\vec{x}^{(k)} - \vec{\xi}\|_\infty$. We say the convergence has **order** $m > 1$ if m is the highest power for which the error quotient $\frac{e_{n+1}}{e_n^m}$ is eventually bounded above. In other words,

$$m = \sup \left\{ \mu \mid \limsup_{n \rightarrow \infty} \frac{e_{n+1}}{e_n^\mu} < \infty \right\}$$

Notice that any finite eventual upper bound will do in this case, and that higher values of m indicate faster convergence. We call the (finite) value of the lim sup of the error quotient the *limiting constant*.

Definition 6.2.44

we say that a method has *linear convergence* when there is no $m > 1$ with a finite lim sup, but $m = 1$ has a limiting constant less than one, i.e.

$$\limsup_{n \rightarrow \infty} \frac{e_{n+1}}{e_n} = c, \quad \text{with } 0 < c < 1$$

If $c = 0$ then the convergence is called *superlinear*.

■ Question 76.

□

Suppose that a sequence recursively defined as $x_{k+1} = g(x_k)$ converges to x_* as $k \rightarrow \infty$. Recall that $x_* = g(x_*)$, i.e. x_* is a fixed point of g . Let g' be continuous in the neighbourhood x_* . Prove the following claims.

(a) If $|g'(x_*)| \neq 0$ then the convergence will be linear with rate $\lambda = |g'(x_*)|$.

(b) If $|g'(x_*)| = 0$ then the convergence will be superlinear.

The special case of $m = 2$ is similarly called *quadratic convergence* since it corresponds to a quadratic bound-relationship like $e_{n+1} \leq ce_n^2$ between errors.

Theorem 6.2.45: Newton converges at least quadratically

Assume $f(\vec{\xi}) = \vec{0}$, assume f is C^2 in a neighbourhood of $\vec{\xi}$, and assume $J_f(\vec{\xi})$ is nonsingular. Then there exists $\varepsilon > 0$ such that if $\vec{x}^{(0)} \in \overline{B_\varepsilon(\vec{\xi})}$, Newton's method is well-defined, converges to $\vec{\xi}$, and the convergence is at least quadratic in the sense of [definition 43](#).

■ **Question 77.**

Group

For Newton's method in one dimension, justify the following steps. What assumptions do we need about f to make it work?

$$(a) \quad 0 = f(x_n) + f'(x_n)(z - x_n) + \frac{f''(\xi_n)}{2}(z - x_n)^2$$

$$(b) \quad x_{n+1} - z = \frac{f''(\xi_n)}{2f'(x_n)}(z - x_n)^2$$

$$(c) \quad \limsup \frac{e_{n+1}}{e_n^2} \leq \left| \frac{f''(z)}{2f'(z)} \right|$$

On the other hand,

Theorem 6.2.46: Superlinear convergence of Broyden's method

Assume $f(\vec{\xi}) = \vec{0}$, f is C^1 in a neighbourhood of $\vec{\xi}$, and $J_f(\vec{\xi})$ is nonsingular. If $\vec{x}^{(0)}$ is sufficiently close to $\vec{\xi}$ and B_0 is sufficiently close to $J_f(\vec{x}^{(0)})$, then Broyden's method converges to $\vec{\xi}$ superlinearly: there exists a sequence $\varepsilon_k \rightarrow 0$ such that

$$\|\vec{x}^{(k+1)} - \vec{\xi}\| \leq \varepsilon_k \|\vec{x}^{(k)} - \vec{\xi}\|.$$

Superlinear convergence means the error ratio $\|\vec{e}^{(k+1)}\|/\|\vec{e}^{(k)}\| \rightarrow 0$: the method accelerates as it approaches the root, but without the guarantee that each step *squares* the error the way Newton does. In practice, Broyden often takes more iterations than Newton near the root, but each iteration is far cheaper, making the total computational cost competitive or superior for large n .

Limitations of Local Convergence. [Theorem 45](#) strictly guarantees fast convergence once we start sufficiently close to a root with an invertible Jacobian. However, it fails to determine:

- which specific root is obtained when several roots exist,
- whether the chosen iterative method securely converges from a given starting coordinate,
- how globally sensitive the algorithmic outcome is to slight perturbations of $\vec{x}^{(0)}$.

To rigorously discuss these global questions, we introduce basins of attraction.

6.2.1 Basins of Attraction

Definition 6.2.47: Basin of attraction

Assume $f(\vec{x}) = \vec{0}$ has distinct solutions $\vec{\xi}_1, \dots, \vec{\xi}_m$. For Newton's method (or any specified iteration mapping), the *basin of attraction* of $\vec{\xi}_j$ is the set:

$$S_j = \{\vec{x}^{(0)} \in \mathbb{R}^n : \vec{x}^{(k)} \rightarrow \vec{\xi}_j\}.$$

Note: The *domain of convergence* of the method is the comprehensive union $\bigcup_{j=1}^m S_j$, inclusive of any points resulting in convergence to non-solution fixed points.

What should we expect these sets to look like? In one dimension, the 'basins' are typically intervals. In two dimensions, basin boundaries can be complicated; near a boundary, small changes in $\vec{x}^{(0)}$ can change the limiting root. In general, the domain of convergence is composed of multiple regions separated by boundaries which can be smooth or fractal curves. Determining the domain of convergence analytically is very difficult, so we use random sampling instead.

Example 6.2.48: A system with four real solutions

Consider the nonlinear system below.

$$f_1(x, y) = x^2 + 4y^2 - 16, \quad f_2(x, y) = xy^2 - 4.$$

- First, draw the graphs of the two functions in DESMOS to make an initial guess of the intersection points.
- Randomly sample a point within a reasonably sized 2D box. Choose $[-6, 6] \times [-6, 6]$ for this exercise, but allow the user to specify the size at runtime.
- Write a program that will repeatedly apply Newton's method to this starting point and keep track of whether the process converges to a specified tolerance (e.g., $\tau = 10^{-8}$.) within a fixed number of iterations. The values of tolerance and number of iterations should be specified as input variables.
- If the algorithm converged, color the starting point by the root to which it converges to (four colors total).
- Repeat this process multiple times with random sampling each time, so that you get a colored picture of the 2D box showing where the starting points converge to.
 - Where are the basin boundaries located relative to the four roots?
 - Do the boundaries appear smooth or fractal?
 - What happens to the iteration count near the lines $x = 0$ and $y = 0$? Why?
- Then make a separate 2D plot. This time, we differentiate the color by the number of steps required to converge to a root. So this will be a continuous gradient, with black for the points

that don't converge in a fixed max number of iterations.

- Looking at the two plots, discuss where the clusters of points are that converge slowest.

■ Question 78.

□

Consider the equation $z^3 = 1$ where $z \in \mathbb{C}$. We know that there are exactly three complex roots. We are going to try to estimate them numerically.

First we convert the equation to a system of equation in real variables.

$$\begin{aligned} f_1(x, y) &= x^3 - 3xy^2 - 1 = 0 \\ f_2(x, y) &= 3x^2y - y^3 = 0 \end{aligned}$$

Then do the following:

- Draw the graphs of the two functions to make an initial guess of the intersection points.
- Assign each of the roots a different color. [Choose Blue, Green, Red.]
- Randomly sample a point within a reasonably sized 2D box. Let's say $[-1, 1] \times [-1, 1]$.
- Write a program that will repeatedly apply Newton's method to this starting point and keep track of whether convergence to a specified tolerance occurred within a set number of iterations. Set tolerance to be 10^{-5} .
- Color the starting point according to the root that it converges to (you can figure it out by calculating ℓ_∞ -distances from the roots, and finding the minimum). Choose a different color depending on the root.
- If the program diverges after the max number of iterations, color it Black. Vary the maximum number of iterations allowed (use $n_{\max} = 5, 10, 20, \text{ and } 100$).
- Repeat this process at least 1000 times with different starting points so that you get a colored picture of the 2D box showing where the starting points converge to.

The set of points that are all the same color will be the basin of attraction for the root associated with that color.

Your output should look similar to [figure 6.2](#).

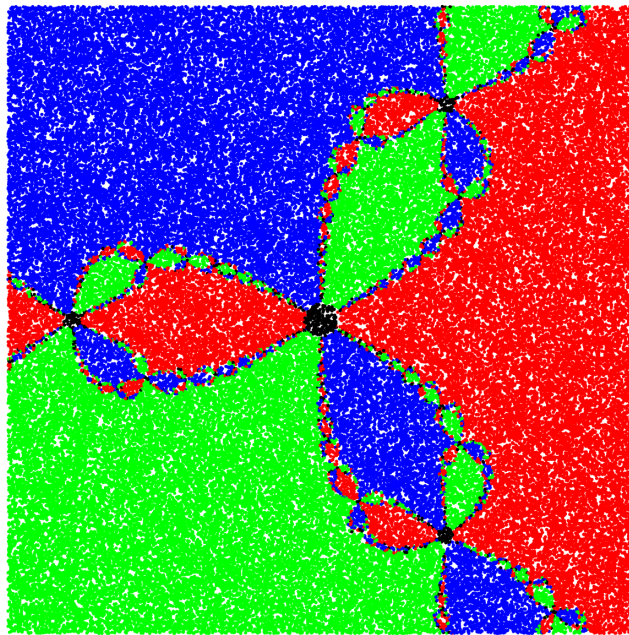


Figure 6.2: Basin of attraction for 10000 random points